

Meridian Health Network

Master Data Management Capstone

Informatica MDM SaaS — Business 360

"One trusted patient. One trusted code. One trusted view of Meridian."

How to Use This Document

- Every milestone is a business problem, not a technical checklist.
- You decide the MDM approach. Business 360 is a candidate tool, not a given answer.
- Each milestone depends on the previous one. Skipping steps compounds the mess.
- Deliverables accumulate. Your final submission is one portfolio: the journey from raw chaos to governed master data.

Dataset Location

```
MDM_Capstone/  
1_Sources/          <- patient extracts, one file per source system  
2_Transactional/    <- appointments, billing, visits, prescriptions, claims, ...  
3_Reference/        <- hospitals, departments, doctors, codes, hierarchies,  
Data_Sources  
4_Master_Goldens/   <- the combined, pre-merge patient file
```

The Four Source Systems

Source	What it is	Staff nickname
EMR_Epic	The flagship clinical record	"Epic"
EMR_Cerner	Second clinical record used by acquired hospitals	"Cerner"
RegistrationDB	The old registration system	"the registrar"
ManualEntry	Hand-typed corrections and walk-in desk entries	"the desk"

Storyline: The Mandate

Meridian Health Network runs 25 hospitals, 21 departments, and 620 doctors across 17 US states. For two decades, every hospital bought or built its own patient systems. Nobody agreed on what a "patient" is, what a "code" means, or which record is correct when two systems disagree. Patient data now lives across the four source systems above.

The new CIO has bought Informatica MDM SaaS (Business 360). She has handed the implementation to you — one team, 10 milestones, one goal. You will start by loading raw files, and finish with a governed, consumable golden record visible in dashboards — just like a real Business 360 rollout.

The milestones build on each other — earlier work is a prerequisite for the work that follows. A later milestone can only be delivered if the ones before it were done right.

Curriculum Coverage Map

Every curriculum topic from the MDM program is exercised somewhere in this journey. The map below shows where each Major Topic is practiced. Use it to make sure your submission proves coverage of the full syllabus — not just the data work.

Major Topic	What you must demonstrate	Milestones
MDM Foundations	Golden record, graph modeling, machine learning (ML) matching, data stewardship, workflows, agile MVP delivery, cloud-native MDM value	1, 5, 6, 7, 8, 10
Business Strategy & ROI	Business drivers, Single Source of Truth business case, ROI — quantified before/after	4, 10
Cloud Data Governance	Axon-type governance: requirements, data domains, source & consumer systems, DQ rules, match rules, trust scores, business relationships, policies, facets, workflows, lineage	2, 3, 5, 6, 8
IDMC Platform	IDMC services, MDM SaaS components, microservices, conceptual architecture, platform data flow	1, 9
Business 360	Purpose and business value of MDM; on-premises vs cloud SaaS deployment comparison	1
Data Modeling	Entities/attributes/relationships, Base Object, XREF & history tables, timeline & state management, tokenization, null handling, reference data, parent-child, staging tables, mapping, clears/clean functions, packages & triggers	3, 6, 9
Hierarchies	Business entities, entity views, identification strategies, search configuration, hierarchy import, provisioning tool administration	7
Match & Merge / Hierarchy Manager & Admin	Exact/fuzzy match rules, XREF/Match/STRP tables, auto & manual merges, entity & relationship hierarchies, hierarchy types, batch jobs & scheduling, provisioning tool UI	5, 7, 9
Data Quality	DQ dimensions (accuracy, completeness, consistency, timeliness), DQ lifecycle (profile → cleanse → validate → govern), cloud DQ services, DQ functions & outputs	2, 4
Data Profiling	IDMC profiling, rule specifications, DQ rules, CLAIRE AI recommendations, rule occurrences, scorecards, CDGC, exception management	2
Advanced Data Profiling	Source mapplets for multi-source profiling, DQ rules on complex sources, CDGC scorecards, advanced mode rule specifications, dictionaries, conditional logic	2, 4
Cleanse Functions	Standardization, validation, conditional processing, Read Database lookups, DQ enforcement with custom libraries / IDQ integration	4

Milestone 1: The Welcome Coffee

Where do we even start?

Milestone 1 of 10

The Business Problem

The first all-hands meeting ends in silence. Nobody knows how many files exist, what's in them, or whether they can even be opened together. The data team has never seen the extracts. Before any strategy discussion, they need a shared, honest snapshot of what Meridian actually has.

Your Decision

Business 360 could help by making the raw files visible as a catalogued data inventory. What do you think? If yes, do it.

Expected Deliverables

- A catalogue of every file, its columns, and row counts
- A first read of data quality per file (rough % of obvious garbage)

- A short memo to the CIO: “here is what we think we have” with no conclusions yet

Hints — Expected Activities & Problem Areas in the Data

- Open and catalogue every file in the four folders — count columns and rows, sample the data.
- Note which files are raw source extracts vs. the consolidated golden patient list.
- The four patient extracts overlap — the same person can appear in more than one source file.
- Some files are near-empty or trivial to scan — flag them, don’t assume they’re fine.
- Business 360 / cloud SaaS: position the value of cloud MDM vs on-premises deployment, and sketch the IDMC services and conceptual data flow you will rely on.

Milestone 2: The Data Profiling

How bad is it, really?

Milestone 2 of 10

The Business Problem

Revenue says “the data is fine”. Clinical says “the data is terrible”. Nobody has ever measured it. The CIO wants a per-source quality scorecard — not opinions. Which system is most complete on name, email, phone, insurance? Which is least? Where are the nulls, the nonsense values, the impossible values?

Your Decision

Business 360/IICS profiling could expose this without touching the data. What do you think? If yes, do it.

Expected Deliverables

- A per-source completeness/quality report
- A clear ranking: which source can be a trust anchor for which fields
- A data profiling summary the CIO can present to the board honestly

Hints — Expected Activities & Problem Areas in the Data

- Build a per-source completeness report for name, DOB, email, phone, insurance — blank % and invalid %.
- Rank the four sources — Data_Sources.csv already carries trust_score and data_quality_score (Epic 90 > Cerner 75 > RegistrationDB 60 > ManualEntry 40).
- Watch for ~130 invalid emails (e.g. not-an-email, user@.com), missing/malformed DOBs, and phone-like strings in email fields.
- Watch for SSN/PII values sitting in the wrong fields (names, occupations, allergies) in 7 records.
- Data Profiling: run profiling / rule occurrences per source, review scorecards and handle exceptions; where available, apply CLAIRE AI column recommendations to shape rules.
- Advanced Data Profiling: use a Source Mapplet to profile the four patient extracts together in one multi-source analysis, and document advanced mode rule specification choices, before deciding trust anchors.

Milestone 3: The Reference Data Mess

Which “Cigna” is which?

Milestone 3 of 10

The Business Problem

Accounting reports show 24 different spellings for a handful of insurance payers. Clinical codes repeat as “n/a”, “None”, “unknown”. Procedure categories disagree between hospitals. Every downstream report starts with “well, it depends on what you mean by...” Reports are untrustworthy because the words are untrustworthy. The board wants one canonical list for payers, blood types, departments, and wards that everyone must use.

Your Decision

Reference 360 exists so reference data is governed before master data is. What do you think? If yes, do it.

Decide this yourself: Decide this yourself: can you even merge patients correctly if you haven't standardised the payers first?

Expected Deliverables

- Standard code lists for the noisy domains
- A lookup/standardisation rule set
- Proof that the canonical lists are being enforced

Hints — Expected Activities & Problem Areas in the Data

- Pick the noisy domains and build canonical code lists (payers, blood type, departments, wards).
- Standardise — e.g. 24 payer spellings (Cigna vs Cigna HealthCare vs CIGNA), Aetna vs Atena.
- Procedure_Codes and the Hierarchy sheet disagree on codes, categories, and names — define which is the system of record.
- Departments.csv / Hospitals.csv labels don't match their IDs when joined to the Hierarchy — the canonical list must reconcile them.
- Reference data as a modeling construct: treat canonical lists as parent-child reference data with ownership, null handling, and versioning rules.

Milestone 4: The Data Quality Rules

Making bad data impossible

Milestone 4 of 10

The Business Problem

The walk-in desk keeps typing phone numbers into the email box, pasting SSN-like strings into name fields, and registering people as "12345". No email means no outreach. No valid phone means no care navigator callback. Bad data keeps arriving because nothing stops it at the door. The CIO wants rules so disciplined that the desk cannot create garbage — and that the existing backlog is quarantined and tracked.

Your Decision

Business 360 and Cleanse functions are designed to standardise, validate, and route bad records. What do you think? If yes, do it.

Expected Deliverables

- Validation rule set (email, phone, name, PII-in-wrong-field, impossible values)
- A quarantine/remediation workflow design
- Counts of what currently fails, by rule and by source

Hints — Expected Activities & Problem Areas in the Data

- Define validation rules: email pattern, phone pattern, value-vs-type, PII-in-wrong-field, impossible numbers.
- Name each rule against the DQ dimensions — Accuracy, Completeness, Consistency, Timeliness — and show the DQ lifecycle (profile → cleanse → validate → govern).
- State the business drivers: duplicate billing, duplicate care, lost outreach — and build the ROI case the rules will protect.
- Cleanse Functions: design standardisation + validation + conditional processing; use Read Database / dictionary lookups to standardise payers and department codes.
- Advanced practice: dictionaries for value replacement, and conditional (low-code) logic instead of brute-force rules.
- Design the quarantine flow — bad records go somewhere visible and get tracked, never silently accepted.
- Watch for 39 first names that are actually numbers like "12345".

- Watch for impossible values — negative billing, negative doctor experience, weight like “heavy” or 650, ratings above 5.
- Watch for future-dated and pre-dated events (DOBs like 2099/1800, discharge before admit, upcoming appointments in the past).

Milestone 5: The Duplicates

Mrs Smith? Which Mrs Smith?

Milestone 5 of 10

The Business Problem

The registration desk files a duplicate ticket every day. Some patients genuinely appear twice — once in Epic, once in the registrar, with a slightly different name. One person has four records under four systems. Another two “patients” share a telephone number but are completely different people, and the system keeps wanting to merge them. Care is being duplicated, billing is paying twice, and no one can prove which records are the same person.

Your Decision

Business 360’s match-and-merge is built for exactly this problem. What do you think? If yes, do it.

Expected Deliverables

- Matching strategy (which keys, which fuzzy logic, what thresholds — your call)
- The merge-consolidation result: count and examples
- Proof you did NOT merge distinct people who merely share a phone

Hints — Expected Activities & Problem Areas in the Data

- Define the match strategy yourself — keys, fuzzy rules, thresholds — and record your false-positive controls.
- Explain how the match and merge process writes XREF / Match / STRP tables and how auto vs manual merges are managed.
- Use machine learning (ML)-driven matching insight (e.g. CLAIRE-style recommendations) to justify your thresholds and match confidence.
- Watch for 70 near-duplicate groups: same phone with different name, email typos, name+DOB lookalikes, abbreviation variants.
- Watch for PID 100 (John/Jon/John S./J. Smith) and PID 200 (Maria/Marie Garcia variants) — each present across four sources.
- Watch for 5 cross-walk patients carrying 3 IDs each (positive / negative / appended).
- ~250 patients share a phone and ~22 share an email — but NOT all of them are the same person. Do not over-merge.

Milestone 6: The Golden Record and Its Parents

Who owns this value?

Milestone 6 of 10

The Business Problem

After merging, someone finally says the dangerous sentence: “so... which source is RIGHT?” For the four-record patient, Epic says one email, the desk says another, the registrar says a third DOB. Meridian needs a golden record for every person, but must prove where each value came from and why that source won. Legacy_HIS contributed records for years and is now inactive — its values must count for less.

Your Decision

Business 360 uses trust scores, source weighting, and survivorship to decide — and it keeps the audit trail. What do you think? If yes, do it.

Expected Deliverables

- The full golden record for the multi-source conflicted patient, with per-attribute justification
- Trust / decay / survivorship policy
- An audit view: “value X came from source Y, chosen for reason Z”

Hints — Expected Activities & Problem Areas in the Data

- Define source weights and trust decay — Legacy_HIS is inactive, so its values must rank lower than active systems.
- Deliver one value per attribute with the audit trail: “value X came from source Y because Z”.
- Data Modeling: design the subject-area model — Entity / Base Object with XREF and history tables; how timeline, state management, and null handling behave; where tokenization applies to PII.
- Graph modeling: represent patient relationships (family, employer, provider) so the “360 view” is a graph, not just a row.
- Watch for multi-source conflicted patients — four systems give different email, DOB, and phone for the same person.
- Watch for manual and registration values conflicting with clinical (Epic) values on the same patient.

Milestone 7: The Hierarchies

Who sits under whom... and who's a ghost?

Milestone 7 of 10

The Business Problem

Operations printed the org structure and someone noticed: insurance plans exist that point at a parent company that doesn't exist; a department is listed under two hospitals; a doctor is registered to a hospital with no such number; and the surgery procedure tree loops back on itself. The hierarchy is the operating map of the network. It's broken.

Your Decision

Business 360's hierarchy configuration and Reference 360's hierarchy import could own this. What do you think? If yes, do it.

Expected Deliverables

- Complete defect inventory per hierarchy (insurance, procedures, hospital-depth, doctor-dept)
- The corrected hierarchy model
- A validation that the hierarchy can be consumed by downstream systems

Hints — Expected Activities & Problem Areas in the Data

- Build a full defect inventory for every hierarchy: insurance, procedure codes, hospital→department, doctor→hospital.
- Configure business entity identification and entity views, set search configuration and default values for the golden record.
- Import the hierarchies, then validate them — this exercises hierarchy import and hierarchy types.
- Watch for Insurance_Provider_Hierarchy — ghost parent, self-reference, inverted tier, inactive parent with active child.
- Watch for Procedure_Codes — circular codes, broken chain to 999, disconnected subtree.
- Watch for the Hierarchy sheet — phantom hospital/department links, duplicate department assignment, unregistered hospital.
- Watch for Doctors.csv — ghost doctor on hospital 999, duplicate/near-zero records, ratings above 5.

Milestone 8: The Guards

Who may touch what — and who approves a new patient?

Milestone 8 of 10

The Business Problem

A nurse almost edited a patient's legal name. An auditor with read-only access deleted a test row. A billing clerk can see SSN-like values that are supposed to be masked from everyone except data stewards. Meridian needs role-based, attribute-level security, plus an approval gate before any new identity enters the golden record — so duplicates get caught before publication, not after.

Your Decision

Business 360 ships preconfigured roles, attribute-level access, and approval workflows. What do you think? If yes, do it.

Expected Deliverables

- A role/permission matrix mapped to the real roles in the story
- Attribute-level masking demonstrated
- An approval workflow with steps, roles, and notifications

Hints — Expected Activities & Problem Areas in the Data

- Define role-based, attribute-level access and map real roles: data steward, provider, billing, auditor, admin.
- Demonstrate attribute-level masking (SSN-like fields) and an approval gate before a new identity is created.
- Decide who may edit what — legal name is steward-only; billing must not see raw PII; duplicates get caught before publish.
- Cloud Data Governance (Axon-type): capture the patient domain — data domains, source & consumer systems, policies, and lineage from source to golden record.
- Governance workflows: map stewardship tasks and approvals as governed workflow steps with owners and notifications.

Milestone 9: The Operation

The desk still has to feed the beast

Milestone 9 of 10

The Business Problem

The golden record is beautiful. But tomorrow the desk will type more garbage, Epic will send updates, and a batch feed will crash because someone renamed a column. The whole pipeline — file in → validate → standardise → match → survive → publish — must run on a schedule, fail loudly, and publish to downstream systems and the patient portal without ever exposing junk.

Your Decision

Business 360's ingestion (file import, mappings, mapplets) and egress (file/API/Salesforce) are the operating heartbeat. What do you think? If yes, do it.

Expected Deliverables

- The end-to-end ingestion design (where rules run, where rejects go)
- The egress contract for at least one consumer
- Proof of a scheduled, traceable flow with failure handling

Hints — Expected Activities & Problem Areas in the Data

- Design the end-to-end flow: file in → validate → standardise → match → survive → publish — and where rejects go.
- Explain the platform data flow: staging tables → mapping → Base Object, driven by packages and triggers.
- Batch jobs: group steps into batches and schedule them (with an external scheduler), following batch group best practices.
- Define at least one egress contract (Billing, patient portal, or a reporting consumer) plus a run schedule.
- Use Data_Sources.csv — refresh_frequency and last_sync_date are meant to drive the schedule and decay logic.
- Feeds will break (renamed columns, malformed rows) — design alerts, reprocessing, and fail-loud behaviour.

Milestone 10: The One Screen

Show the board the truth

Milestone 10 of 10

The Business Problem

The board meeting is in a week. The CIO wants one screen that proves the program works: source volumes and which systems feed what; match rates and duplicate backlog trending down; data quality percentage per domain; hierarchy health; and a live, searchable golden record — a single patient looked up and shown as one trustworthy row.

Your Decision

Business 360's search, operations insights, dashboards, and the golden-record view are the finale. What do you think? If yes, do it.

Expected Deliverables

- A dashboard/visual design covering the five dimensions above
- A demonstrated golden-record lookup end to end
- A final "before vs now" story for the board

Hints — Expected Activities & Problem Areas in the Data

- Build one screen across five dimensions: source volumes, match rate / duplicate backlog trend, DQ % per domain, hierarchy health, and golden-record search.
- Demonstrate a real lookup — one patient, 4 raw rows, merged to 1 governed row with lineage.
- Produce a before/after comparison with real numbers: duplicate %, orphans, quality %, hierarchy defects.
- Data-driven applications: expose the golden record to a consuming app / API and show the business value achieved (Single Source of Truth, ROI story).
- Agile delivery: show the MVP approach — what was delivered in each iteration across the journey and how feedback shortened the path to value.

Submission Portfolio

- Executive summary — what Meridian had, what you built, the numbers that prove it
- Per-milestone deliverables — each milestone, the business problem, your approach and why, the result, and how you verified it
- The golden record showcase — at least one patient traced from 4 raw records to 1 governed record with full lineage
- Before/after metrics — duplicates, orphans, quality %, hierarchy defects, all showing movement
- Lessons learned + what you would do at a real enterprise scale

Evaluation Criteria

Dimension	Weight
Problem understanding (business-first framing)	20%
Data analysis and evidence (you can find the problem in the data)	25%
Correctness of MDM approach per milestone	25%
Quality of implementation/deliverable	20%
End-to-end story and communication	10%

A rollout that skips Milestone 3 (reference data) cannot deliver Milestone 5 (merge). A rollout that skips Milestone 5 cannot deliver Milestone 10 (one screen). The flow is the point.